B - Count Adjacent Cells

分值：200 分

Problem Statement

There is a grid with H rows and W columns. The cell at the i-th row from the top and the j-th column from the left is denoted as cell (i, j).

有一个 H 行 W 列的网格。从上往下第 i 行、从左往右第 j 列的格子记为 (i, j)。

Cells (x_1, y_1) and (x_2, y_2) are said to be **edge-adjacent** when $|x_1 - x_2| + |y_1 - y_2| = 1$.

当 $|x_1 - x_2| + |y_1 - y_2| = 1$ 时，我们称格子 (x_1, y_1) 和 (x_2, y_2) **边相邻。**

For every cell, find the number of cells that are edge-adjacent to it.

对于每个格子，求与它边相邻的格子数量。

Constraints

- $1 \le H, W \le 50$
- All input values are integers.
- $1 \le H, W \le 50$
- 所有输入值都是整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

H W

Output

Output the answer in the following format:

按照以下格式输出答案：

```
$x_{1,1}$ $x_{1,2}$ $\cdots$ $x_{1,W}$
$x_{2,1}$ $x_{2,2}$ $\cdots$ $x_{2,W}$
$\vdots$
$x_{H,1}$ $x_{H,2}$ $\cdots$ $x_{H,W}$
```

Here, $x_{i,j}$ represents the number of cells that are edge-adjacent to cell (i, j).

其中，$x_{i,j}$ 表示与格子 (i, j) 边相邻的格子数量。

Sample Input 1

```
4 5
```

Sample Output 1

```
2 3 3 3 2
3 4 4 4 3
3 4 4 4 3
2 3 3 3 2
```

The cells edge-adjacent to cell $(1, 5)$ are cells $(1, 4), (2, 5)$, for a total of two cells.

与格子 $(1, 5)$ 边相邻的格子是 $(1, 4), (2, 5)$，共 2 个。

The cells edge-adjacent to cell $(2, 3)$ are cells $(1, 3), (2, 2), (2, 4), (3, 3)$, for a total of four cells.

与格子 $(2, 3)$ 边相邻的格子是 $(1, 3), (2, 2), (2, 4), (3, 3)$，共 4 个。

The cells edge-adjacent to cell $(4, 2)$ are cells $(3, 2), (4, 1), (4, 3)$, for a total of three cells.

与格子 $(4, 2)$ 边相邻的格子是 $(3, 2), (4, 1), (4, 3)$，共 3 个。

Sample Input 2

```
1 1
```

Sample Output 2

```
0
```

There are no cells edge-adjacent to cell $(1, 1)$.

没有与格子 $(1, 1)$ 边相邻的格子。

Sample Input 3

```
12 8
```

Sample Output 3

```
2 3 3 3 3 3 3 2
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
3 4 4 4 4 4 4 3
2 3 3 3 3 3 3 2
```